

- 1 (a) error that does not have a fixed magnitude or direction B1
- or
- error that scattered around the true value.
- or equivalent statement
- (b) (i) $13 \cos 35^\circ = 10.6$ or 11 m s^{-1} B1
- (ii) $v_{\max} = 13 \times 1.1 \cos 34^\circ = 11.9 \text{ m s}^{-1}$ B1
- or
- $v_{\min} = 13 \times 0.9 \cos 36^\circ = 9.47 \text{ m s}^{-1}$
- uncertainty = $11.9 - 10.6 \approx \pm 1 \text{ m s}^{-1}$ B1
- or $10.6 - 9.47$ or $\frac{11.9-9.47}{2}$
- $(11 \pm 1) \text{ m s}^{-1}$ B1
- uncertainty given to 1 sf and value given to the same precision as uncertainty
- 2 (a) No net force B1